



*Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)*

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## ***DESIGN AND MODELING OF SHELL AND TUBE HEAT EXCHANGER***

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The domain of the Project:  
AUTOCAD AND SOLIDWORKS

Team Mentors (and their designation):

**Mr. Mairala Chinnarao**

(Design Engineer)

(Sedin Technologies)

Team Member:

Mr. VINAYAK KABBURI

Period of the project

July 2026 to August 2026



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## Declaration

The project titled “DESIGN AND MODELING OF SHELL AND TUBE HEAT EXCHANGER” has been mentored by **Mr. Mairala Chinnarao**, organised by SURE Trust, from March 2026 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Submitted By  
Mr. Vinayak Kabburi

Project Mentor:  
Mr. Mairala Chinnarao

Prof. Radhakumari  
Executive Director & Founder  
SURE Trust



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### ***Executive Summary***

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- This project focuses on the **design, development, and 3D modelling of a Shell and Tube Heat Exchanger using SolidWorks**. A shell and tube heat exchanger is one of the most widely used types of heat transfer equipment in industries such as power generation, chemical processing, petroleum, food processing, refrigeration, and manufacturing. The primary purpose of the equipment is to transfer heat between two fluids at different temperatures while keeping the fluids physically separated.
- The main objective of this project was to develop a **complete and accurate three-dimensional CAD model** of a shell and tube heat exchanger and to gain a better understanding of its construction, component arrangement, and assembly process. The work began with a study of the **basic principles of heat transfer and heat exchanger operation**, followed by an understanding of different heat exchanger configurations, major components, and applications. Preliminary design considerations and sizing calculations were also carried out to determine suitable dimensions for the major components.
- The modelling process was carried out using **SolidWorks**, where the individual components were created as separate 3D parts. The major components modelled include the **cylindrical shell, tube bundle, tube sheets, channel heads, nozzles, flanges, support saddles, lifting lugs, and other external fittings**. Appropriate dimensions and geometric relationships were considered to ensure proper fitting and alignment of the components during assembly.
- After completing the individual part models, all components were integrated into a **complete assembly using SolidWorks assembly features and mates**. Dimensional verification was carried out to identify possible interference, alignment, and fitting issues, and necessary modifications were made to improve the accuracy of the overall model. Material properties were also assigned to the components based on their intended application and operating requirements, considering factors such as **strength, corrosion resistance, thermal properties, durability, and manufacturability**.
- Overall, the project provided practical knowledge of **mechanical design, CAD modelling, assembly techniques, dimensional control, material selection, and heat exchanger construction**. The final outcome is a complete 3D CAD assembly of a Shell and Tube Heat Exchanger developed in SolidWorks, demonstrating the application of mechanical engineering design principles to the development of industrial heat transfer equipment.



## ***Introduction***

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A **Shell and Tube Heat Exchanger** is one of the most commonly used industrial heat-transfer devices due to its simple construction, reliability, and suitability for a wide range of operating conditions. It consists of a bundle of tubes enclosed within a cylindrical shell, allowing heat to be transferred between two fluid streams while keeping the fluids physically separated. Depending on the application, one fluid flows through the tubes while the other flows through the shell side. These heat exchangers are widely used in industries such as **power generation, chemical processing, petroleum, refrigeration, food processing, and manufacturing.**

The design and modelling of a heat exchanger require proper consideration of its major components, dimensions, arrangement, and material properties. In this work, **SolidWorks** was used to convert the functional arrangement of the heat exchanger into a detailed three-dimensional CAD model. The modelling process included the creation of individual components such as the **shell, tubes, tube sheets, nozzles, flanges, support saddles, lifting lugs, and other external fittings.**

The scope of the work mainly focused on **component modelling, assembly development, dimensional checking, and material assignment.** The individual components were modelled with appropriate dimensions and subsequently assembled to obtain the complete heat exchanger configuration. The final CAD model provides a clear representation of the equipment and its construction, while the work primarily emphasizes **mechanical design and CAD development rather than experimental thermal performance testing.**



### ***Project Objectives***

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- To study the **construction, working principle, and basic configuration** of a Shell and Tube Heat Exchanger.
- To understand the **major components and their functions** in a shell and tube heat exchanger.
- To perform **preliminary sizing and dimensional calculations** required for developing the CAD model.
- To create accurate **individual 3D CAD models** of the major heat exchanger components using SolidWorks.
- To model important components such as the **shell, tube bundle, tube sheets, nozzles, flanges, support saddles, lifting lugs, and external fittings**.
- To assemble all the individually modelled components into a **complete and properly aligned heat exchanger assembly**.
- To verify the **dimensions, alignment, positioning, and assembly relationships** of the components.
- To identify and minimize possible **interference and fitting issues** during the assembly process.
- To assign **suitable engineering materials** to the completed CAD model based on the intended application and component requirements.
- To develop a **complete 3D CAD representation** that can be used for design visualization, documentation, and further engineering analysis.



## Methods / Technology Used

- Study of shell and tube heat exchanger fundamentals and component arrangement.
- Preliminary dimensional and sizing calculations.
- Parametric 3D part modelling in SolidWorks.
- Assembly modelling using mates and alignment constraints.
- Dimension verification and correction.
- Material assignment to the completed assembly.

- ## Explanations

The following three images are CAD views of the work completed in SolidWorks. Each view highlights a different part of the design and shows the progression from the internal tube arrangement to the complete assembled equipment.

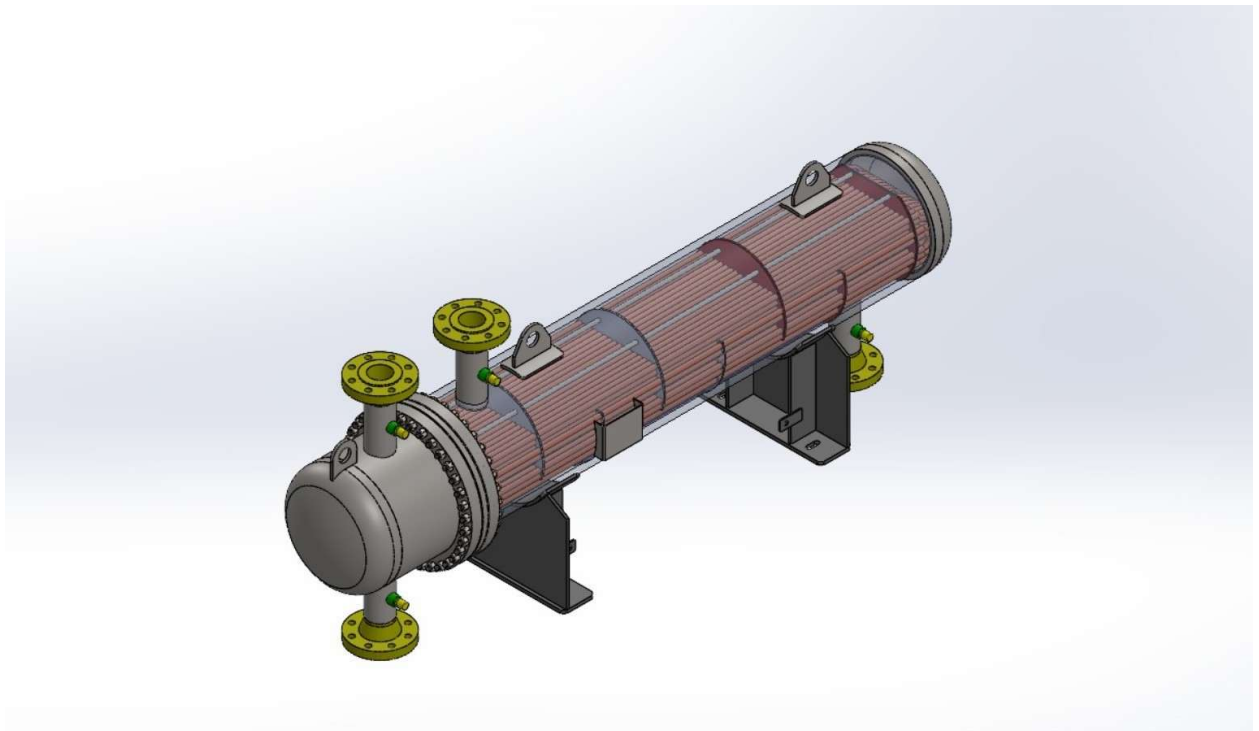


Figure 1: Complete Shell and Tube Heat Exchanger Assembly



**Explanation:** This view shows the completed overall assembly. The main cylindrical shell contains the tube bundle, while the model also includes the front-end cover, flanged connections, nozzles, lifting lugs and support structures. The view is useful for checking the overall component arrangement, alignment and integration of the individual SolidWorks parts. It represents the final assembly stage of the CAD work.

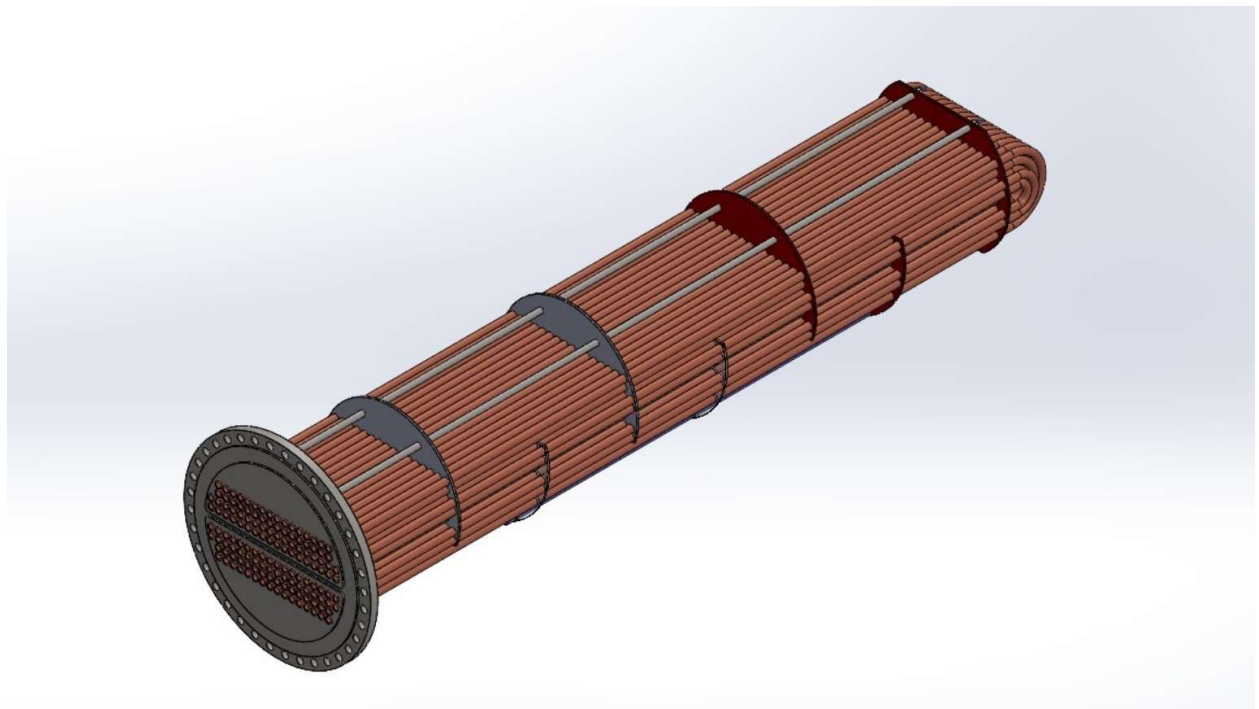


Figure 2: Tube Bundle and Internal Arrangement

**Explanation:** This image focuses on the internal tube bundle. A large number of parallel tubes are arranged within the shell, with supporting plates/baffles visible at intervals. The tube ends are collected at the tube-sheet/end region. This view is important because the tube bundle is the main heat-transfer area of the exchanger. It also demonstrates the CAD work involved in arranging many repeated tube components accurately inside the shell.



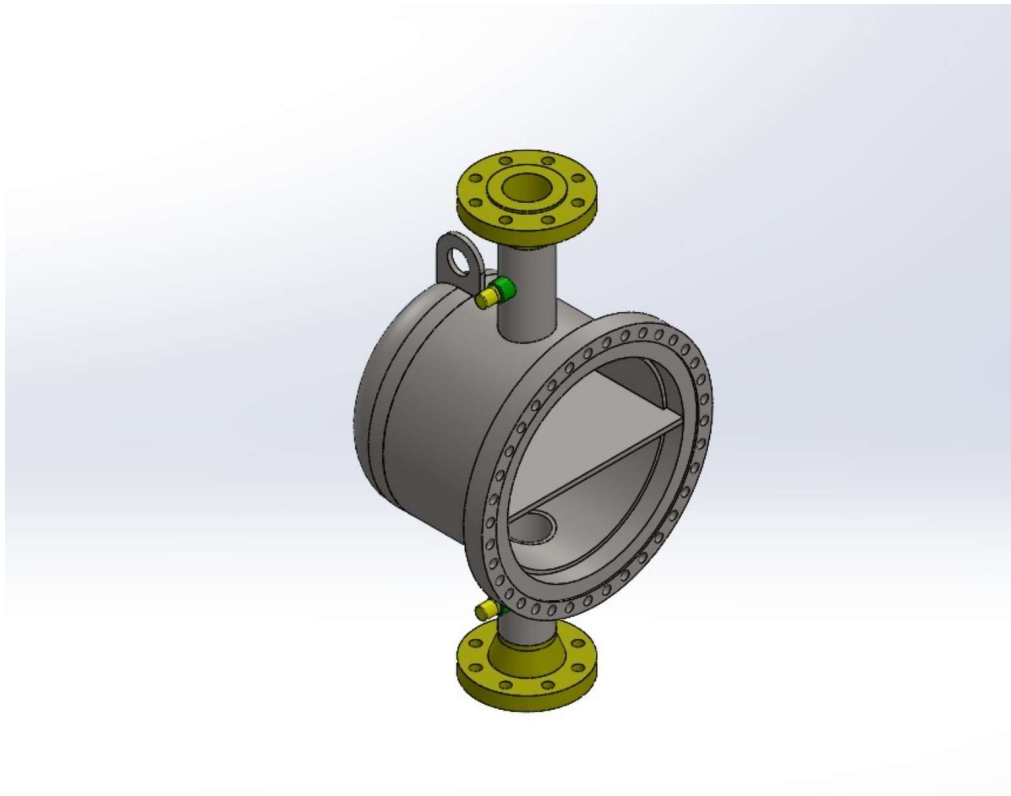


Figure 3: Shell, Nozzles and Flanged Connections

**Explanation:** This view shows the shell-side structure separately from the tube bundle. The cylindrical shell, large bolted flange, top and bottom nozzles, lifting lug and internal support/baffle arrangement can be clearly seen. This view helps verify the external connections and the relationship between the shell opening, flanges and internal components before or during the final assembly.



## Material Selection and Explanation

Material assignment was completed in the CAD model to represent engineering materials and to make the assembly more realistic. The exact grade of every component is not stated in the supplied project document, so the explanations below distinguish between what is visible/recorded and typical engineering choices.

| Material/Component                    | Engineering Justification & Benefits                                                                                                       |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stainless Steel – Shell</b>        | High mechanical strength, excellent weldability, and good corrosion resistance, making it suitable for the main pressure-containing shell. |
| <b>Copper Alloy – Tubes</b>           | Superior thermal conductivity, which improves heat transfer and helps optimize convective heat exchange efficiency.                        |
| <b>Carbon Steel – Tube Sheets</b>     | Robust structural support with good machinability, allowing accurate manufacturing of tube holes and maintaining proper tube alignment.    |
| <b>Stainless Steel – Nozzles</b>      | High rigidity under fluid forces and good resistance to environmental and process-related corrosion.                                       |
| <b>Carbon Steel – Flanges</b>         | High pressure-handling capability, structural integrity, and good resistance to thermal cycling under operating conditions.                |
| <b>Nitrile Rubber – Gaskets</b>       | Good compression recovery, temperature resistance, and effective sealing between mating components to prevent fluid leakage.               |
| <b>Carbon Steel – Support Saddles</b> | High structural yield strength and heavy load-bearing capacity, providing stable support for the heat exchanger assembly.                  |



## Project Architecture

- Main cylindrical shell body
- Tube bundle and internal tube arrangement
- Tube sheets / end connections
- Nozzles and flanged connections
- Support saddles
- Lifting lugs and external fittings
- Final assembly with material assignment

## Modelling and Assembly Process

- Studied the basic construction and identified the major components required for the CAD model.
- Performed preliminary calculations and selected the required dimensions.
- Created individual part models for the shell and associated components.
- Modelled nozzles, flanges, support saddles, lifting lugs and other external fittings.
- Created and arranged the tube bundle within the shell assembly.
- Assembled the created models using appropriate assembly relationships.
- Verified dimensions and corrected alignment where required.
- Completed the final assembly and assigned materials to the components.

## Results

The major components were successfully modelled and assembled into a complete shell and tube heat exchanger CAD model. The final assembly represents the shell, tube bundle, end connections, nozzles, supports and external fittings. The completed model was also assigned suitable material representations for engineering visualization and documentation.



Shell and tube heat exchangers are widely used in **process industries, power plants, chemical and petrochemical industries, HVAC systems, oil and gas facilities, food processing, refrigeration, and other thermal engineering applications**. They play an important role in efficient heat recovery, cooling, heating, and temperature control of industrial fluids.

The design and modelling of such equipment are important aspects of **mechanical and process equipment engineering**. Developing a 3D CAD model provides practical exposure to component design, assembly techniques, dimensional control, material selection, and engineering documentation. Learning to model a shell and tube heat exchanger using SolidWorks therefore provides useful preparation for careers in **mechanical CAD, equipment design, manufacturing, and industrial engineering**, while also improving understanding of real-world engineering equipment and its construction.



### ***Learning and Reflection***

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## Technical Learning

- Improved understanding of shell and tube heat exchanger construction.
- Improved ability to convert component requirements into parametric SolidWorks models.
- Gained experience in creating complex assemblies from multiple individual parts.
- Improved skills in dimension verification and assembly alignment.
- Learned the importance of component sequencing and reference geometry in CAD modelling.
- Gained practical exposure to assigning engineering materials in SolidWorks.

## Overall Experience

The work provided hands-on experience in taking a mechanical system from concept study to a complete CAD assembly. It improved confidence in SolidWorks part modelling, assembly creation, component integration and engineering documentation.



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## Future Scope & Conclusion

### Conclusion

The objectives were achieved by studying the shell and tube heat exchanger configuration, completing preliminary design work, modelling the required components and integrating them into a complete assembly. The final model was checked for dimensional consistency and materials were assigned. The project strengthened practical skills in mechanical CAD and SolidWorks assembly modelling.

### Future Scope

- Perform detailed thermal design calculations and heat-transfer analysis.
- Carry out CFD analysis to study fluid flow and pressure drop.
- Perform structural analysis of selected components.
- Optimize tube arrangement and geometric parameters for improved performance.
- Prepare detailed manufacturing drawings and a bill of materials.



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